
Flood-Induced Dynamic Road Network and Evacuation Island Analysis in Taliaokeng Creek Watershed

資管三乙 412402024 黃姿婷

資管三乙 412402256 林湘凌

資管三乙 412402488 簡伊辰

1. INTRODUCTION

In recent years, extreme rainfall events have become more frequent and have increased the risk of urban flooding. When roads are flooded, residents may have difficulty reaching evacuation shelters within a limited time, which can lead to the formation of evacuation islands. Therefore, understanding evacuation accessibility during flood events is important for disaster prevention and emergency planning.

GIS and network analysis provide useful tools for evaluating accessibility under disaster scenarios. However, many traditional network analyses assume that all roads remain accessible and do not consider road disruptions caused by flooding. To better reflect real-world conditions, this study applies a dynamic road network approach in which road accessibility changes according to flood depth.

This study uses the Taliaokeng Creek Watershed as the study area. By integrating road networks, flood inundation data, evacuation shelters, and population data, we identify evacuation islands and estimate the affected population under flood conditions. The results can support future flood mitigation and evacuation planning.

2. STUDY AREA

The study area is the Taliaokeng Creek Watershed, which includes Xinzhuang District and Shulin District in New Taipei City and Guishan District in Taoyuan City (Figure 1). The area is highly urbanized and contains residential, commercial, and industrial land uses.

Because of its dense population and extensive road network, flooding may affect residents' ability to reach evacuation shelters. Therefore, this watershed was selected to evaluate evacuation accessibility and evacuation island distribution under flood conditions.

The study uses road network data, flood inundation maps, evacuation shelter locations, and population grid data to analyze the impact of road disruptions on evacuation services.



Figure 1. Study area of the Taliaokeng Creek Watershed.

3. DATA COLLECTION

This study collected diverse spatial datasets to construct a dynamic road network and evaluate evacuation accessibility within the Taliaokeng Creek Watershed. The spatial datasets incorporate a digital elevation model (DEM), a micro-topographic sink layer, a manually extracted flood barrier dataset, and an authoritative digital road network. All raster datasets were clipped and aligned to the specific study area comprising Xinzhuan, Shulin, and Guishan districts.

To effectively evaluate emergency preparedness within the Taliaokeng River Watershed, two primary datasets are required: **social exposure data** and **emergency shelter locations**. High-resolution demographic data is essential to identify where residents live and establish a baseline of the populated clusters. Simultaneously, verified shelter data pinpoint exactly where safe evacuation zones are located. By integrating and aligning these two datasets within a GIS environment, the model can accurately analyze whether local residents can successfully access emergency shelters during a disaster.

3.1 Terrain analysis

3.1.1 DEM data

The DEM data of Taoyuan City and New Taipei City (Figure 2) were obtained from the Government Open Data Platform of Taiwan. The data have a spatial resolution of 20 meters. This DEM served as the

primary terrain dataset from which slope and topographic depression analyses were derived to understand the terrain relief, flow direction, and potential water accumulation zones within the Ta-Liao-Keng Creek watershed.

Taoyuan and New Taipei City Digital Elevation Model

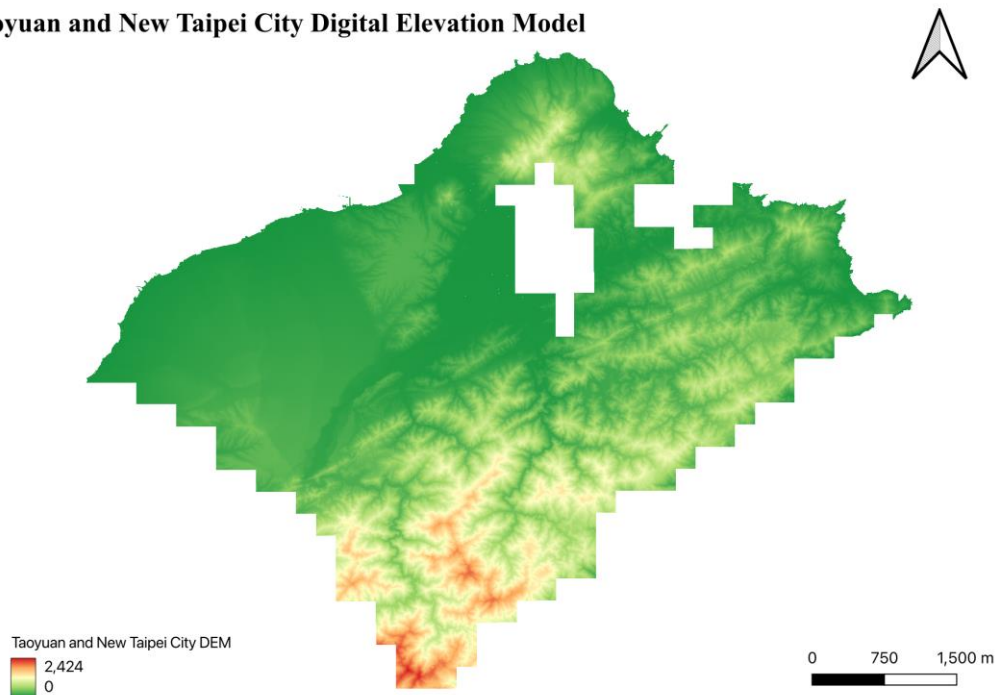


Figure 2. Taoyuan and New Taipei City Digital Elevation Model

3.1.2 Slope data

Slope data were derived from the DEM using the slope analysis tool in QGIS. The slope values were calculated based on elevation differences between neighboring cells and expressed as percentage slope to reflect surface relief and runoff transport conditions. As shown in Figure 3.

Terrain Slope Analysis of Taliakeng River Watershed and Districts

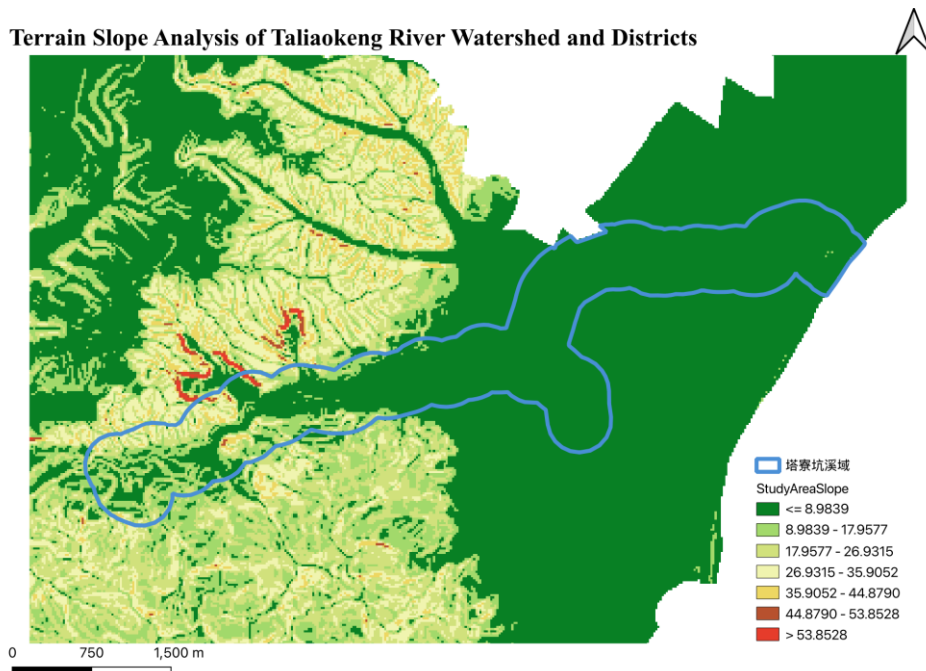


Figure 3. Terrain Slope Analysis of Taliaokeng River Watershed and Districts

3.1.3 Topographic depressions

Topographic depressions were identified by comparing the original DEM with the depression-filled DEM. During rainfall events, surface runoff tends to accumulate in these depressions, making them more susceptible to waterlogging and flooding. Therefore, identifying topographic depressions helps reveal potential flood-prone locations that may not be apparent from elevation or slope data alone. As shown in Figure 4.

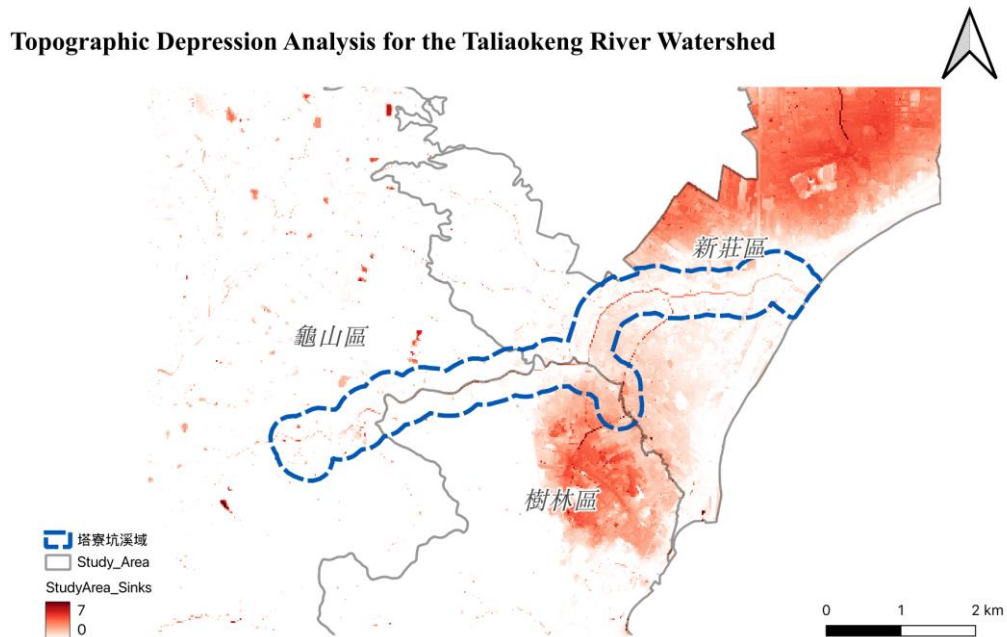


Figure 4. Topographic Depression Analysis for the Taliaokeng River Watershed

3.1.4 Flood potential maps

Flood potential maps were obtained from the Flood Disaster Safety Project Information Service Website. The flood inundation datasets represent simulated flooding scenarios under different rainfall intensities and durations. As shown in Figure 5.

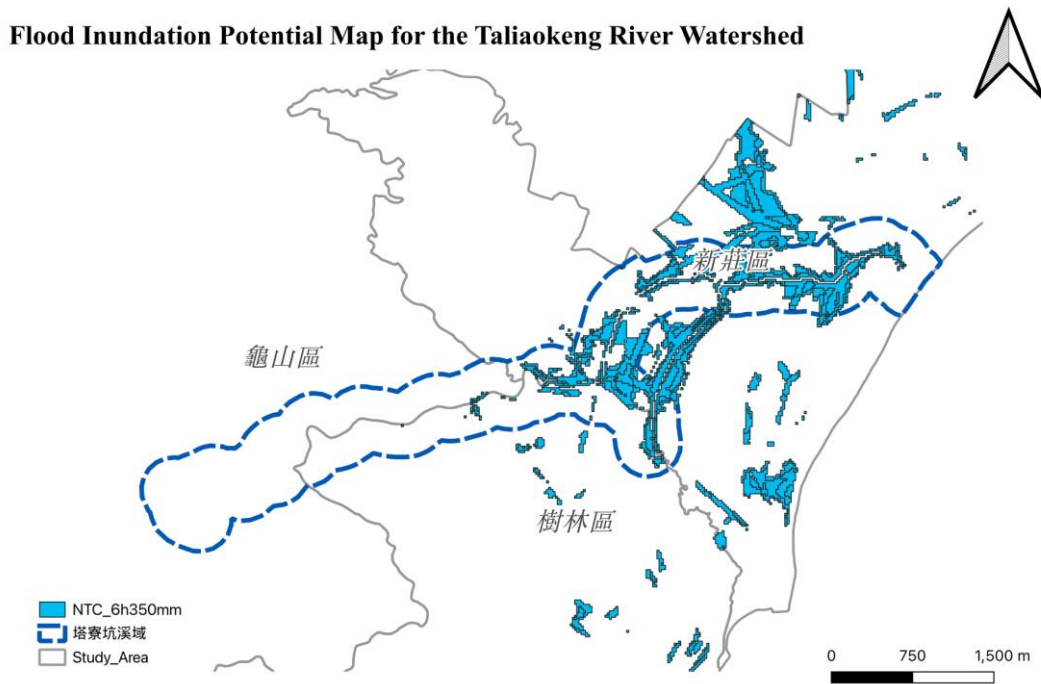


Figure 5. Flood Inundation Potential Map for the Taliaokeng River Watershed

3.2 Transport network inputs

Regarding transport data, this study used road centerlines derived from OpenStreetMap as the base road network, then performed topological cleaning to ensure correct connectivity and node structures within the network, facilitating subsequent impedance adjustment and accessibility analysis. Next, the road data were spatially overlaid with flood depth data by matching the geographic position of each road segment with the corresponding inundation raster cells, so that each segment could inherit a representative flood depth attribute. In practice, this overlay process allowed the flood depth values intersecting each road segment to be extracted and assigned to the road layer, forming the basis for dynamic cost calculation under different flood conditions. In addition, barrier points were established for high-risk locations such as underpasses and historical flooding hotspots to represent impassable nodes during a disaster, thereby simulating road closure effects under flooding scenarios. As shown in Figure 6.

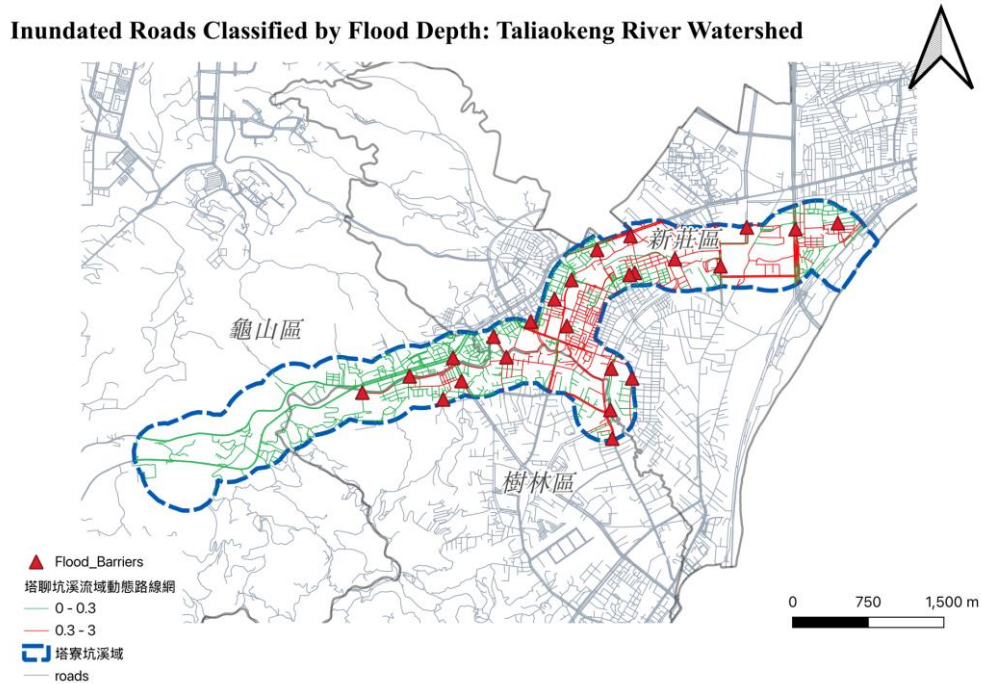


Figure 6. Inundated Roads Classified by Flood Depth: Taliaokeng River Watershed

3.3 Social exposure inputs

Regarding social exposure data, this study utilized demographic statistics obtained from the Socio-Economic Geographic Information System (SEGIS) of the Ministry of the Interior to establish the baseline of the populated residential areas. The raw demographic dataset was obtained in a tabular CSV format which lacked inherent spatial geometry. To visualize and analyze the exposure distribution, an attribute-based table join operation was executed within the GIS environment. This process linked the CSV population attributes to a corresponding 100m × 100m vector grid polygon shapefile using the unique grid identification codes (CGRID_ID). The successfully joined layer serves as the core social exposure baseline, identifying exactly which grid cells represent populated residential clusters and allowing the model to test whether these specific residents can access emergency shelters.

To ensure temporal consistency across the trans-boundary Taliaokeng River Watershed, the integrated grid layer systematically synchronizes the December 2025 dataset for both New Taipei City and Taoyuan City. During the preprocessing stage, these separate municipal datasets were merged into a single layer. Data type standardization was subsequently conducted to convert the population attribute fields from string formatting into numeric data types, followed by a zero-fill operation to address null values, thereby preventing arithmetic errors during downstream spatial calculations.

In addition, emergency shelter registries were retrieved from the Open Data Platforms of both New Taipei and Taoyuan City Governments. Since the official public registries only provided textual descriptive addresses without explicit spatial coordinate attributes, a meticulous manual geocoding and digitization workflow was conducted. Each shelter address was individually cross-referenced, verified, and located via Google Maps to extract its precise geographic coordinate pair. These manually digitized point features were then imported into the GIS environment and underwent coordinate transformation to the TWD97 / Taiwan

TM2 projection (EPSG:3826) to ensure strict geometric alignment with the prepared road network and population grids. The associated attribute tables were subsequently filtered to retain only actively designated evacuation facilities, establishing the verified destination nodes required for downstream accessibility routing and spatial difference analysis.

4. METHODS

4.1 Spatial Sampling of Flood Data

Flooding potential maps from NCDR/WRA for Taoyuan and New Taipei City are ingested via Rasterio and merged into a unified spatial dataset retaining GRIDCODE values. A GeoPandas spatial join then intersects the topologically cleaned road network against the merged flood polygons, exploding each road segment into multiple rows where it traverses flood zones of varying depth. To enforce a conservative, safety-first approach, the algorithm selects the maximum GRIDCODE among all intersecting polygons per segment, extracting a single representative row to form `final_roads`. A micro-terrain correction is subsequently applied by cross-referencing GRIDCODE values with DEM-derived slope and topographic sinks — segments within zero-slope depressions receive elevated flood risk, while steep-slope segments are adjusted downward to reflect surface runoff. Finally, the corrected GRIDCODE values are mapped to actual water depth attributes (`Flood_Depth`) and exported as `roads_with_flood_depths.shp`, forming the spatial foundation for downstream disaster routing and service area analyses. The flowchart is presented in Figure 7.

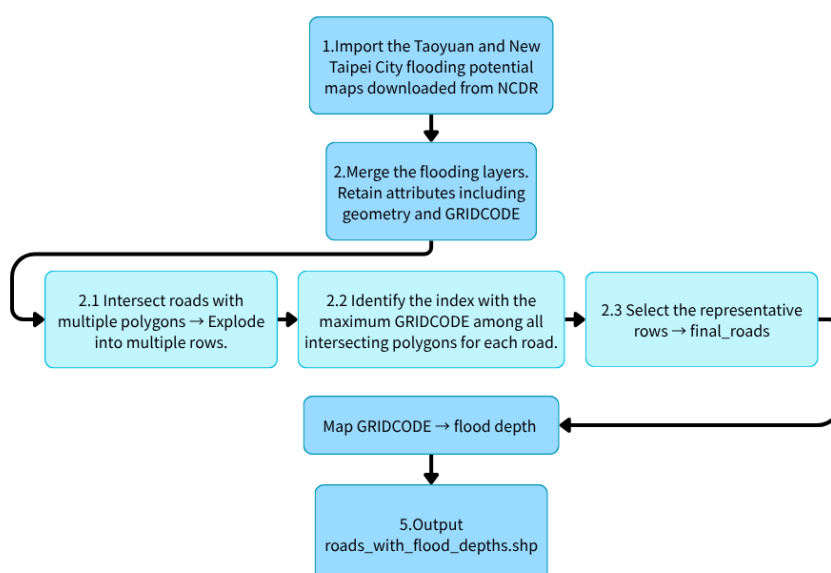


Figure 7. Spatial Sampling of Flooding Data Workflow

4.2 Smart Road Network and Dynamic Impedance

To simulate the impact of flooding on road accessibility, flood depth information and high-risk road segments were integrated into the road network. Road segments were classified as impassable when the average flood depth or maximum flood depth exceeded 0.3 m, or when they were identified as high-risk road segments. An extremely high impedance value was assigned to represent road closure during flood events, while the original walking time was retained for passable segments. The workflow of the dynamic impedance assignment is shown in Figure 8.

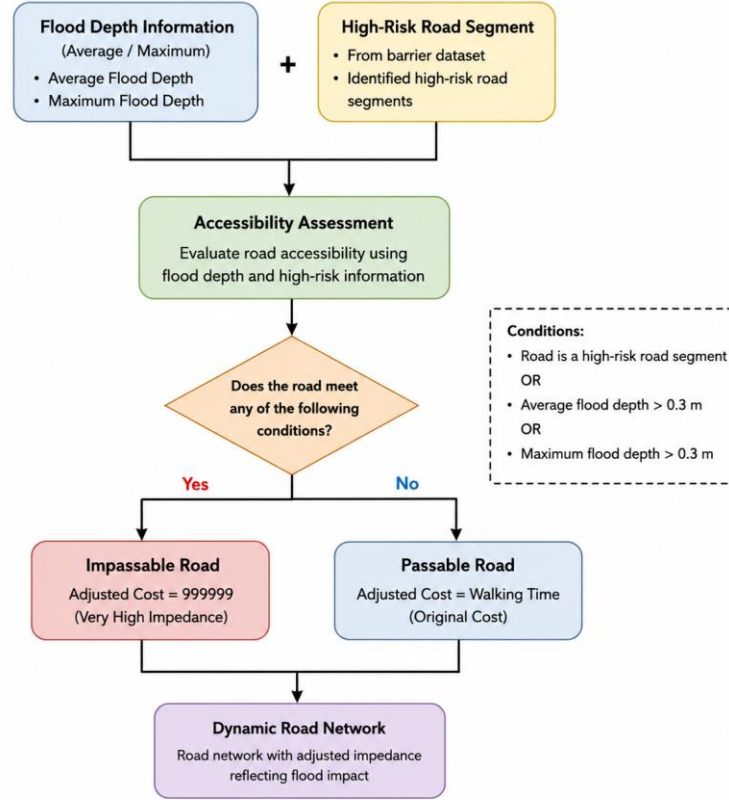


Figure 8 Dynamic Impedance Assignment

4.3 Evacuation Catchment and Network Isolation Analysis

Regarding the network routing and containment analysis, the preprocessed road network layer (*roads*) was first filtered to enforce disaster-induced physical barriers. Segments identified with severe inundation depths—where the evaluated travel impedance reached the critical impassability threshold (i.e., $Adjusted_C \geq 999999$)—were systematically excluded from the network topology graph to simulate realistic disaster road closures. Utilizing this flood-adjusted road graph, a network service area analysis was executed, setting the manually digitized emergency shelters as the origin destination nodes. The allocation routing strategy was configured based on the shortest geometric distance, applying a strict travel cost constraint of 1,200 meters to mathematically represent a standard 15-minute evacuation walking threshold for an adult pedestrian under emergency conditions.

To transform the resulting linear service networks into a continuous accessible geographic zone, a 100-meter spatial buffer operation was applied to the generated route lines, followed by a dissolve execution to merge overlapping polygons into a seamless safety catchment layer. Subsequently, to isolate the residential

communities completely cut off from emergency resources, a spatial Difference overlay operation was performed. This geometric exclusion subtracted the dissolved 15-minute safety catchment polygons from the prepared SEGIS population grid layer (established in Section 3.3). The remaining vector polygons successfully delineated the precise geographic boundaries of the "Evacuation Islands"—representing populated residential clusters trapped without viable walking paths to any designated shelter—forming the topological input for downstream area-weighted demographic aggregation. The flowchart is presented in Figure 9.

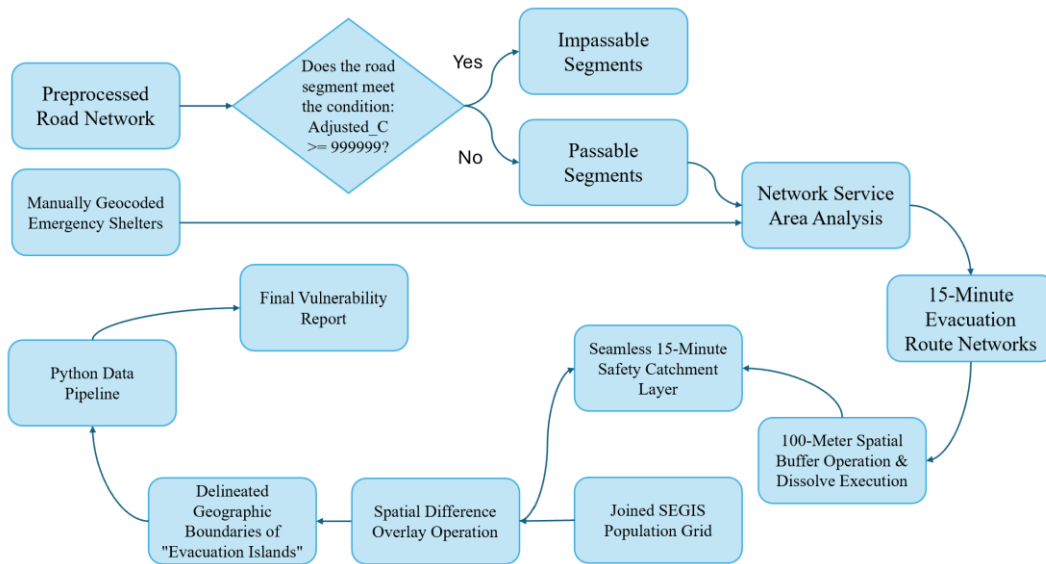


Figure 9. Evacuation Catchment and Network Isolation Analysis Flowchart

4.4 Geometric overlay and area-weighted demographic aggregation

To accurately quantify the exact population entrapped within each isolated zone, a spatial intersection overlay (`gpd.overlay(..., how='intersection')`) was executed between the delineated "Evacuation Islands" and the merged SEGIS population grid layer. This spatial operation cross-references the geometric boundaries, fragmenting the $100\text{m} \times 100\text{m}$ population grid cells into smaller, irregular polygons where they intersect with the island boundaries. Because a population grid cell might be partially severed by an evacuation island boundary, assigning the entire grid's population to a single fragment would lead to significant statistical overestimation. To enforce strict demographic accountability, an area-weighted interpolation method was implemented. The actual geometric surface area of each fragmented polygon (`fragment_area`) was dynamically calculated and divided by the baseline area of a full grid cell (10,000 square meters) to derive an area proportional ratio (`area_ratio`). The total population attribute of the fragment—which systematically combined the numeric values of the New Taipei and Taoyuan data streams—was then multiplied by this ratio to compute the adjusted, geographically representative population (`adj_total`) for each unique fragment.

Following the geometric adjustment, a programmatic data pipeline developed in Pandas was deployed for statistical aggregation. The fragmented records were grouped by their unique island identifier (`island_id`), and the adjusted populations were summed to calculate the cumulative number of trapped

residents per evacuation island. To ensure data integrity for reporting, the aggregated population sums were rounded to the nearest whole number and cast into integer data types. Finally, the structured results were exported in two distinct formats to serve separate analytical tasks: the tabular statistics were saved as a CSV file (*Final_Vulnerability_Report.csv*) using UTF-8-sig encoding for immediate table preparation, while the integrated geometric boundaries coupled with the calculated population attributes were saved as a clean GeoJSON file (*final_map_with_data.geojson*), forming the spatial foundation for advanced cartographic styling and label rendering within QGIS. The analytical sequence is presented in Figure 10.

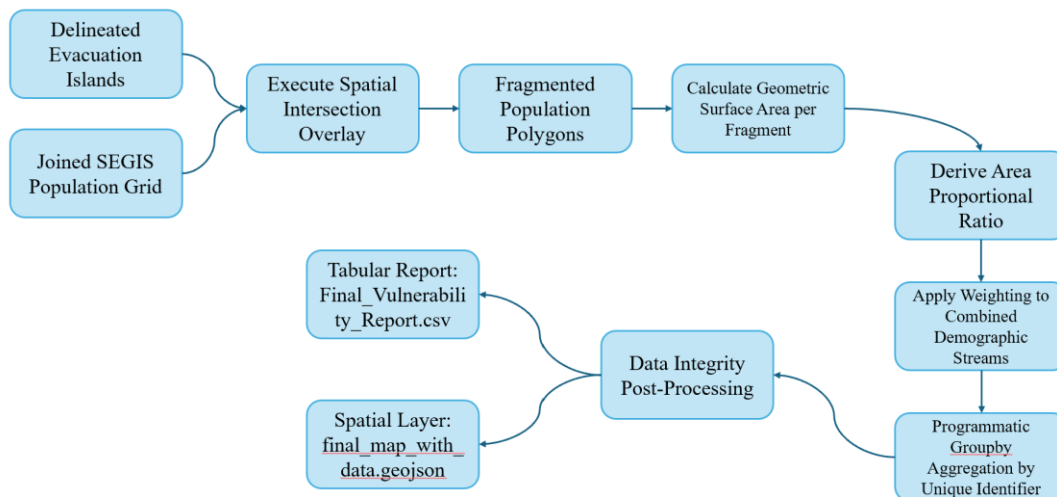


Figure 10. Geometric Overlay and Area-Weighted Demographic Aggregation Flowchart

5. RESULTS AND DISCUSSIONS

5-1 Before-and-After dynamic disruption Road Conditions

Before dynamic disruption of the road network (Figure 11): All roads within the watershed are displayed in a uniform gray color, representing a fully connected road network under normal conditions, where all road segments are passable.

After dynamic disruption of the road network (Figure 12): Roads are classified into two categories based on flood depth:

- Green (0–0.3 m): The water depth does not exceed the threshold, and pedestrian movement remains possible; therefore, the Adjusted_Cost is set equal to travel time in minutes.
- Red (0.3–2 m): The water depth exceeds 0.3 m, and the road segment is considered impassable; thus, the Adjusted_Cost is assigned a value of 999999.

This classification shows that flooding does not affect the entire watershed uniformly, but instead creates a spatially uneven pattern of disruption. Shallow flooding mainly reduces comfort or walking efficiency, while deeper flooding causes complete loss of accessibility on affected road segments. As a result,

the road network changes from a continuous system into a partially disconnected system, where some routes remain usable and others become impassable. The adjusted road costs therefore provide a practical way to simulate dynamic disaster conditions in the network model: under normal conditions, route planning is based on ordinary walking time, but under flooding scenarios the model automatically excludes dangerous segments by assigning extremely high costs. This makes it possible to identify critical bottlenecks, disconnected areas, and alternative routes that remain available during a flood event, and it also shows that flood depth changes not only the physical appearance of the road network but also its functional accessibility.

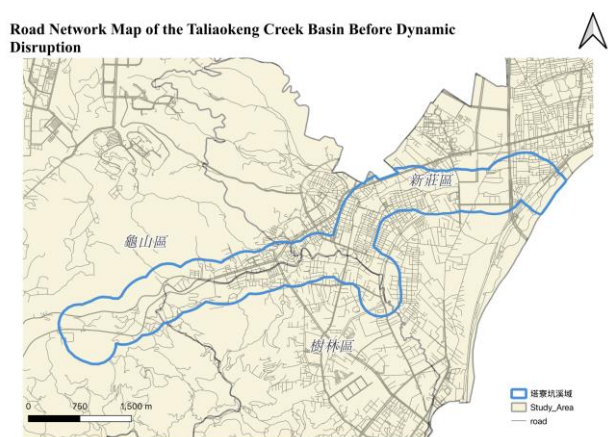


Figure 11. Road Network Map of the Taliaokeng Creek Basin Before Dynamic Disruption

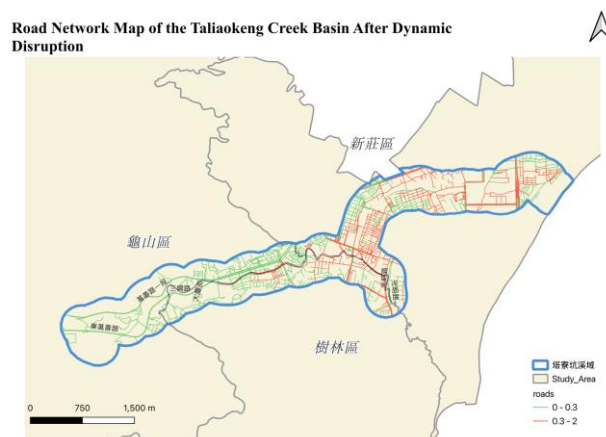


Figure 12. Road Network Map of the Taliaokeng Creek Basin After Dynamic Disruption

5-2 Evaluation of the 15-minute evacuation service catchments

The 15-minute evacuation service catchment network (Figure 13) visually encapsulates the functional reachability of designated emergency shelters under simulated flood constraints. This network layer was generated using the QGIS "Service Area (From Layer)" algorithm, executing a shortest-path strategy with a strict geometric travel cost constraint of 1,200 meters—mathematically calibrated to represent a standard 15-minute adult walking threshold at an average speed of 80 meters per minute. Crucially, rather than relying on complex dynamic impedance parameters within the routing engine, this analysis was performed on a road network layer (roads) that had already been subjected to a hard spatial filter ("Adjusted_C" < 999999). This preprocessing step physically omitted all impassable segments (where Adjusted_C = 999999 due to severe flooding exceeding 0.3 meters), forcing the routing engine to treat disrupted intersections as absolute physical dead ends.

As a result, the map displays a highly uneven, fragmented linear topography rather than a uniform concentric buffer. Main unflooded arterial roads exhibit elongated, continuous peach-pink routing lines that extend far into the watershed, creating a localized spatial illusion of extensive safety coverage along major transport corridors. In stark contrast, peripheral local streets and neighborhood alleys terminate abruptly, truncating at the exact coordinates where they intersect with the hidden, flooded segments. This structural pattern reveals a critical disaster resilience insight: in a localized watershed crisis, evacuation accessibility is

strictly governed by topological connectivity rather than proximity or geometric distance. A residential grid located physically close to an open main thoroughfare remains functionally isolated if the minor intersecting roadways feeding into that thoroughfare are hydrologically severed, validating the necessity of incorporating physical road-omission filters to expose true blind spots in emergency shelter placement.

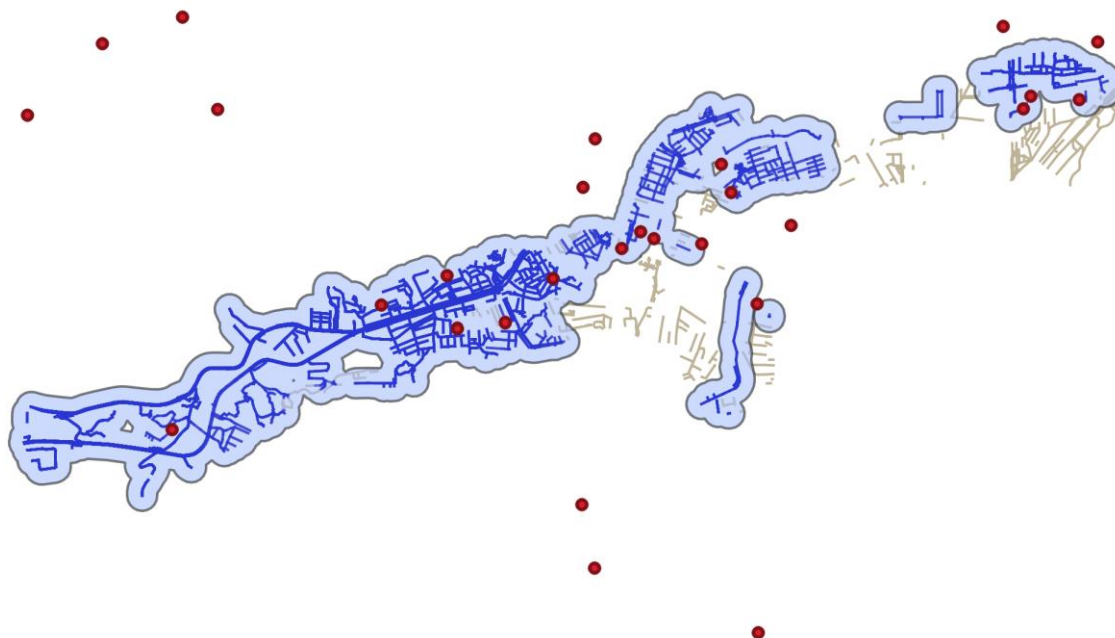


Figure 13. 15-minute evacuation service catchment network

5-3 Evaluation of the 15-minute evacuation service catchments

The final evacuation islands map (Figure 14) represents the ultimate geometric output of the spatial exclusion pipeline, explicitly isolating the residential grid cells cut off from all emergency shelter resources under the flood scenario. This layer was generated by first applying a 100-meter spatial buffer operation to the 15-minute linear route networks (detailed in Section 5-2) and executing a dissolve command to merge overlapping boundaries into a seamless, continuous safety catchment polygon. To extract the trapped population clusters, a spatial Difference overlay was performed within the GIS environment, utilizing the integrated SEGIS population grid as the input layer and the dissolved safety catchment polygon as the cutting mask. This geometric subtraction systematically erased all populated grid zones that fell within a safe walking distance of any shelter, leaving behind a highly fragmented patchwork of vector polygons that define the precise boundaries of the "Evacuation Islands."

The resulting map completely bypasses generalized concentric circular patterns, revealing instead a complex topology comprising 387 distinct, isolated grid fragments distributed unevenly across the Taliakeng River Watershed. These islands are visually rendered with custom ID labels and area-weighted population counts (e.g., Island #86 with 1,525 residents, Island #99 with 1,271 residents) calculated via the downstream GeoPandas pipeline. The spatial distribution demonstrates that significant pockets of high-density residential blocks—particularly in low-lying zones flanking the fragmented secondary road networks—are functionally stranded. The visual empirical evidence in Figure 8 underscores that proximity to a rescue facility does not guarantee safety; due to the hard omission of flooded road infrastructure, multiple

demographic clusters are trapped just hundreds of meters away from an accessible zone, validating the spatial difference method as a rigorous tool for identifying hidden systemic vulnerabilities in urban flood emergency routing.

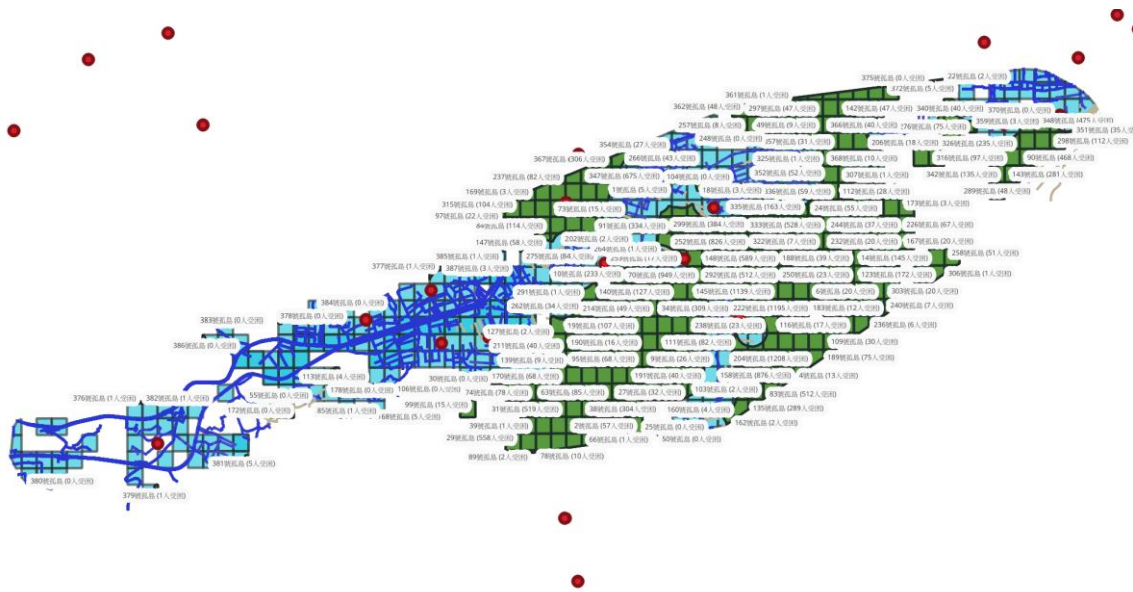


Figure 14. final evacuation islands map

5-4 Tabular Analysis of Evacuation Island Demographics

The final evacuation islands map (Figure 8) represents the ultimate geometric output of the spatial exclusion pipeline, explicitly isolating the residential grid cells cut off from all emergency shelter resources under the flood scenario. This layer was generated by first applying a 100-meter spatial buffer operation to the 15-minute linear route networks (detailed in Section 5-2) and executing a dissolve command to merge overlapping boundaries into a seamless, continuous safety catchment polygon. To extract the trapped population clusters, a spatial Difference overlay was performed within the GIS environment, utilizing the integrated SEGIS population grid as the input layer and the dissolved safety catchment polygon as the cutting mask. This geometric subtraction systematically erased all populated grid zones that fell within a safe walking distance of any shelter, leaving behind a highly fragmented patchwork of vector polygons that define the precise boundaries of the "Evacuation Islands."

The resulting map completely bypasses generalized concentric circular patterns, revealing instead a complex topology comprising 387 distinct, isolated grid fragments distributed unevenly across the Taliakeng River Watershed. These islands are visually rendered with custom ID labels and area-weighted population counts (e.g., Island #122 with 1,525 residents, Island #280 with 1,319 residents, and Island #86 with 1,212 residents) calculated via the downstream GeoPandas pipeline. The spatial distribution demonstrates that significant pockets of high-density residential blocks—particularly in low-lying zones flanking the fragmented secondary road networks—are functionally stranded. The visual empirical evidence in Figure 8 underscores that proximity to a rescue facility does not guarantee safety; due to the hard omission of flooded road infrastructure, multiple demographic clusters are trapped just hundreds of meters

away from an accessible zone, validating the spatial difference method as a rigorous tool for identifying hidden systemic vulnerabilities in urban flood emergency routing.

Rank	Isolated Island	Total Stranded Population (n)
1	122	1525
2	280	1319
3	274	1232
4	86	1212
5	204	1208
6	179	1207
7	222	1195
8	129	1139
9	145	1139
10	323	1117

Figure 15. Final_Vulnerability_Report.csv(top10)

Overall, the discussion confirms that evacuation islands are shaped by the combined effect of terrain, road topology, and land-use distribution. Low-lying road sections and underpasses act as the main isolation triggers, while key connector roads determine whether neighborhoods remain reachable during flooding. The results therefore support the use of dynamic road impedance and spatial difference analysis as an effective way to identify hidden vulnerability zones and prioritize mitigation measures.

6. CONCLUSION

This study successfully integrated flood depth, terrain, road network, and barrier points to establish a smart road network model capable of reflecting dynamic road changes under the influence of flooding. The results demonstrate that low-lying areas and high-risk flood susceptibility zones significantly increase the probability of road disruption, while the incorporation of barrier points further enhances the simulation of disaster-induced road closures and network fragmentation. Through this methodology, static road costs are converted into dynamic impedances that vary with flooding scenarios, making the network analysis more reflective of real-world disaster situations.

We developed a dynamic impedance model by integrating flood depth information and high-risk road segments into the road network to simulate changes in road accessibility under flood conditions. The dynamic cost assignment effectively represents flood-induced road disruptions and provides a more realistic basis for network accessibility analysis. The resulting dynamic road network serves as an important foundation for evacuation service area analysis and emergency evacuation planning.

This study evaluated network accessibility and spatial isolation by executing a 15-minute evacuation catchment analysis and a geometric difference overlay against populated demographic grids. The integration of topological service areas with population data effectively represents the localized pocket effects and

structural blind spots induced by severe inundation. The resulting delineation of evacuation islands serves as a scientific framework for emergency evacuation planning and urban flood risk management. Specifically, these insights enable disaster mitigation agencies to transition from conventional uniform resource allocation to data-driven prioritization, guiding the strategic pre-positioning of emergency supplies, the optimization of shelter accessibility, and the targeted reinforcement of critical road connections within highly vulnerable isolated zones.

7.DATA RESOURCES

- **20m DEM**
[https://www.tgos.tw:443/MDE/VirtualDir_TC/Product/9574dbb5-ca32-4ea3-aaf7-a9f582b9357b/分幅_新北市_20MDEM\(2025\).zip](https://www.tgos.tw:443/MDE/VirtualDir_TC/Product/9574dbb5-ca32-4ea3-aaf7-a9f582b9357b/分幅_新北市_20MDEM(2025).zip)
[https://www.tgos.tw:443/MDE/VirtualDir_TC/Product/ba0fcc26-f82e-4345-bfc4-4785574d9014/分幅_桃園市_20MDEM\(2025\).zip](https://www.tgos.tw:443/MDE/VirtualDir_TC/Product/ba0fcc26-f82e-4345-bfc4-4785574d9014/分幅_桃園市_20MDEM(2025).zip)
- **Flood potential maps**
https://www.dprcflood.org.tw/DPRC/file/Flooding%20potential/shape_file_3/02_shp_%E6%96%B0%E5%8C%97%E5%B8%82.zip
- **Transport network**
<taiwan-latest-free.shp.zip>
- **Grid-based Population Statistics of New Taipei City**
<https://segis.moi.gov.tw/STATCloud/QueryInterfaceView?COL=2vFT9a7InFKKJ4bEswgZ%252bw%253d%253d&MCOL=zDdKDtyFdkF9rM9ivS%252f7NA%253d%253d>
- **Level-6 Statistical Grid Map of New Taipei City**
<https://segis.moi.gov.tw/STATCloud/QueryInterfaceView?COL=N1ktpFcU90QpSFpQrAILMA%253d%253d&MCOL=orvI5Shm0MunXUKzw3t6Og%253d%253d>
- **Evacuation Shelters of New Taipei City**
<https://data.ntpc.gov.tw/datasets/25e439ab-49e7-4e5e-85ce-a25c13fd2770>
- **Grid-based Population Statistics of Taoyuan City**
<https://segis.moi.gov.tw/STATCloud/QueryInterfaceView?COL=FgWzYGZAawZjj07RQMDOtA%253d%253d&MCOL=7KiGQG2NeRFnErVxB5JFnQ%253d%253d>
- **Level-6 Statistical Grid Map of Taoyuan City**
<https://segis.moi.gov.tw/STATCloud/QueryInterfaceView?COL=Vq67ELKQCnKuiO9MSqtKMg%253d%253d&MCOL=EJKSS5u4xwH2xzfK7F21xA%253d%253d>
- **Evacuation Shelters of Taoyuan City**
<https://opendata.tycg.gov.tw/datalist/247820e9-0bb6-4ff9-a34e-f1df72d8b296>